

CDAC

Jo jo interview mein questions puche woh yaha add kardo, help ho jayegi sabki !!

Asked to chinmay:

- Q1. your part in project
- Q2. All Port Numbers
- Q3. if ssh is on port number 5000 then what will be the command to connect that machine ?
- Q4. How to check which ports are open on a Linux machine?
- Q5. What is a digital certificate ?
- Q6. Everything related to the project.
- Q7. How did you created SSL certificates for your project (Flow of the creating RootCA > SubCA > User Certs)
- Q8. I have a database server in my project so they asked me How did you OS harden your Database server?
- Q9. What is NTP ?
- Q10. I did use the FortiGate firewall and Enterprise Edition Nagios XI in my project so they asked details about it, where did you get those? How did you configure it? What makes it different from other alternative tools available ?
- Q11. DHCP ? Its working and its service name in Linux.
- Q12. DNS ? DNS records (A, AAAA, PTR, MX, TXT, CNAME)
- Q13. Scenario: In your project you have a Web server in the DMZ. An attacker tries to get access to it using SSH. How can you prevent it ?
- Q14. Linux Run Levels
- Q15. What is SELinux and three modes of SELinux?

Asked to Prathmesh:

- 1. What is a CA certificate?
 - 2. What is SIEM?
 - 3. What's the difference between Ansible and Puppet?
 - 4. HTTPS vs HTTP
 - 5. How data is encrypted in HTTPS
 - 6. SSL vs TLS?
 - 7. What is a Docker Image
 - 8. How to create a Docker image
 - 9. Deployment process of a container
 - 10. Command to run the container
 - 11. What were the different counter measures taken for your secure system /server? (RDX)
 - 12. What is input validation?
 - 13. What is a honeypot and what's its purpose?
 - 14. How can you say that a website is safe?
 - 15. SAST vs DAST
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Asked to Manu:

1. DHCP process ?
 2. Process for generating Certificates ?
 3. Port Numbers (dns,dhcp)
 4. What do you mean by virtualization ?
 5. OSI model and TCP/IP Model ?
 6. Different way for securing the server ?
 7. Iptables Vs FirewallD ?
 8. How to change file permissions ?
 9. Questions on nagios and its plugins
 10. Software Hardening and Hardware Hardening ?
 11. What do you mean by clustering ?
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Asked to THE Utkarsh Mishra (Company - CDAC Pune)

Round 1:

Q1. Explain your project.

Q2. What is the significance of 2 Master Nodes in your Kubernetes cluster?

Q3. 2 Master Nodes nodes cannot work in a cluster at a time only one works, so how does your cluster know when a master is down so it can shift to another master?

Q3 (**Altered as I couldn't give a straight answer to that lol**): How are all the nodes communicate with each other?

Q4. What have you configured in OPNsense firewall?

Q5. How is OPN sense firewall is different from other industrial firewalls used in companies like cisco, etc.?

Q6. What is NTP?

Q7. How does all of the nodes in your kubernetes cluster synchronize time?

Q8. What will happen if the time of nodes aren't synched in a kubernetes cluster?

Q9. How to configure NTP?

Q10. Are you willing to relocate?

Q11. Are you okay with Night Shifts?

Q12. Are you okay with Project manager role?

Round 2:

Q1. I can see your college project titled "*Real-time yoga pose detection system using OPenCV and MediaPipe*" please explain this project.

Q2. Assume you have 4 VMs running on AWS cloud, one VM is going down due to unavailability of resources, such as high RAM or CPU usage, how will you find out what is being used more?

Q2.1. Let's say you have a monitoring server's GUI dashboard, what will you check there?

Intermediate Dialogue - *"Mf if you are sick like having a feaver you won't just gulp a paracetamol!? First you will check for the pulse and all so like wise here a whole VM has gone down WHAT WILL YOU CHECK!!"* ***Said the interviewer in an ager intensified to ne***

Q2.2 What are the commands for seeing the CPU and RAM usage?

Q3. What is NVlink?

Q4. *As you know the license for Windows Server is acquired depending on number of cores, so by seeing the license how will you tell for how many cores is that license applicable?*

Looking at my obvious wtf idk face he asks...

Q4.1. Do you know the license is acquired on the basis of the number of cores on the server?

Q4.2 (pseudo). IF you would have known that my next question would have been that if you have a license of 24 cores and you have 70 cores on you then how will you manage the rest of the cores?

****THE F HE IS SAYING (obviously couldn't believe on my ears anymore)****

Q5. You have 24 cores Intel i9 CPU, and you can see all the cores aren't being utilized evenly, some cores are showing high usage what will you check and how will you troubleshoot this?

Q6. What is the port for HTTPS? **(After barking "protocol" instead of "port" in his question for 2 times straight.)**

Asked to UMAR

1. Difference between containerization and virtual machines?
2. What is stored in /etc/passwd?
3. What is stored in /etc/shadow?
4. If we delete user information from /etc/passwd then user will be able to login or not?
5. How to recover root password?
6. What are the parameters in recovering root password?
7. What was your role in project
8. What is NTP?
9. Linux Run Levels?

Asked to Rishi

1. What was your contribution to the Project?
2. Why do we need a backup server ?
3. What is RPO & RTO, since you have implemented the Backup Server?
4. Difference Between IDS/IPS?
5. Difference Between Hashing & Encryption ?
6. What is a Zero day exploit? and how will you prevent it?
7. How many layers does your project have ?
8. Write the command for the VLAN setup?

Asked to aakanksha

1. What was your contribution to the project?
2. What is ntp?
3. Have you worked on any webserver?
4. Linux run level?
5. Port no (http,https,dns,dhcp,graphana)?
6. If there are any changes in git repo link (like any letter in the link is changed for ex . capital C ka small c kiya ya kuch bhi letter small se capital ya capital se small kiya) to what will happen?
7. Diff between authentication and authorization?
9. What is ftp?

1.How much will you rate yourself on linux, on a scale of 5?

2.Linux command for:

- To see open ports on the system. Netstat -tuln
- FIPS (cryptography related) command in linux.
- To see open files on the system. lsof

3.Explanation of project and tools used in the project?

4.Protocol used by GrAsked to Pawankumar:afana for fetching logs from loki server.

5.Difference between Encryption and Hashing and its use cases.

6.OWASP top 10, prevention for Broken Access Control.

7.Linux booting process.

8.Linux run levels.

9.Google uses 2 Factor Authentication or 3 factor authentication ?

10.Tell your personal contribution in the project?

Asked to Neha:

Most of the questions were based on project

1. What is siem? For what purpose is it used?
2. Syslog port number?
3. How to configure syslog server?steps?
4. What is the role of system administrator?
5. List web servers that you know?
6. How to harden the web server?
7. How to make your website secure/https?
8. Difference between encryption and hashing? In which scenario they are used?
9. If any server does not support wazuh agent then how to transfer its logs to wazuh?
10. OSI layers?protocols in each layer?
- 11.what is NTP? There is a new replacement in the market for NTP ,do you know about it?
12. Two utilities used to schedule jobs in linux?
13. What is https ? how to make a website secure?
14. What is a digital certificate?process of generating certificates?

GD round topics

1. How to tackle unemployment in developing countries
2. What is more important in job .. money or job satisfaction?
- 3.impact of covid 19 on globalisation

Yash tech me jisko jo quetions puche vo add kr do

1. Use of grafana
 2. What is RDBMS
 3. Mysql and mariadb relational or non-relational database
 4. What is authentication and authorization
 5. Why used mariadb in project
 6. OSI model layers
 7. Write mysql query
 8. Explain jenkins pipeline from project
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TENERITY

Asked to Sankalp

First round

1. what is kubernetes?
2. whats is a pod?
3. what is deployment?
4. what is a service?

5. how was your services communicating with each other? where you have defined address of these?
6. what roles have you created?
7. explain full vpc arcitechture?
8. why do nat gateway needs elastic ip?
9. how will you solve merge conflict?
10. what is build?
11. explain full ci/cd pipeline?
12. what roles have you created in IAM?
13. what policies does these roles include?
14. why does eks needs these roles?
15. What is jenkins?
16. You are not able to connect to your ec2 how will you solve this?
17. Their is a table which shows you cpu utilization of different machines, I want to get the names of those machines with cpu utilization more than 70%, write a script?
18. What ci/cd tools you have used?
19. Do you know argo cd?
20. Explain a docker file?
21. Explain full ci/cd process?
22. How does pods communicate with each other?
23. How to make sure that a job is done on a particular slave in jenkins?
24. How to make sure that pods are being scheduled at particular node?

Second round

25. How did you included Prometheus and grafana in your cluster?
26. What is ssh? How does it works?
27. How agile helps in devops?
28. What are your 5 weaknesses and strengths?
29. How much time will it take you to read a documentation of 150 pages?
30. Lets say there is a table in a file I want to see 1st and 3rd row, write a script? (cut -d 1,3)
31. How much time will you need to solve this scripting problem problem if I give you internet?
32. For how long can you sit on this chair? On what bases can you say this?
33. If i have code in my pc and i want to deploy it on ec2, how can i do it?

Asked to **THE UTKARSH MISHRA** (Company- IFM)

1. Tell us about yourself?
2. Wow! How did you build computers?
3. How are your skills beneficial for our company requirements?
4. Which Operating Systems have you worked on?
5. When and how did you heard about Kali linux?

6. Which attacks have you performed?
7. How did you made a trojan?
8. How did that trojan worked?
9. Now that you have a good knowledge about Red teaming how will this knowledge be beneficial on the other side of the line? (On Blue team)
10. If your colleague has received a suspicious mail how will you check if that mail is malicious or not?
11. Let's say your colleague clicked on the link and downloaded that malicious file now what are the next steps to mitigate this?
12. Let's say you are in some other country or computers are powered through a centralized power source so you cannot pull out the plug directly, now what you will do?
13. How will you know where all the data is going during the time of attack?
14. If there are 50 computers how will you disconnect each machine from the connection?
15. After this incident has taken care of what are the next steps?
16. Do you have any questions?

Asked to THE *now Lull 🐼* UTKARSH MISHRA (Company- Tenerity)

(Only one round because after witnessing rejection from IFM for which I studied last night I was already feeling a bit down and these bastards even though they've selected the boys already were going full on fuckin with me, whooping my ass and I had zero knowledge of their part.)

1. Explain your project
2. Have you used any cloud based services in your project?
3. Do you have knowledge of any cloud based services?
4. Which services have you worked on in AWS?
5. What is a VPC?
6. If VPC works on a private network how does it connect to internet?
7. What is Public Network and Private Network?
8. What is Natting?
9. How natting works in VPC?
10. What is a subnet?
11. Do you know what is a CIDR?
12. *writes on a god forsaken paper* 10.10.0.0/16, What is it's subnet mask?
13. If I change this IP to *goes on to overwrite on same shit he wrote just moments ago* 10.10.0.1/24, calculate how many IPs you will get in this?
14. What is the formula for calculating number of hosts?
15. Which services have you worked on in AWS? *This question was asked 3 times to me, bro has serious ADHD issues*
- 16.
17. What type of storage is S3? Like is it a Object Storage or Blob Storage?
18. How do you connect to an EC2 instance?
19. Let's say you have a storage attached to your EC2 instance and you lost that instance's key, then how will you retrieve your files?

20. Okay what part have you worked on your project?
21. Did you setup your Kubernetes cluster On-Prem or on Cloud?
22. Did you use Mini-Kube?
23. How did you install the kubernetes?
24. What are the main components of Kubernetes?
25. What is etcd?
26. How did you deploy your application?
27. What is a Deployment?
28. What does the deployment yaml file contains?
29. How do you push your application image to docker hub?
30. Write a Dockerfile. *forgot to write EXPOSE*
31. How do you build this image?
32. How do you run your application container now? *As I answered this I realised I forgot to write EXPOSE, their smile made me instant regret T.T*
33. What is the purpose of using exec command in docker?
34. How much do rate yourself in Linux out of 10?
35. Explain me the Linux File System structure.
36. How will I create a group?
37. If I have a file and it's owner is root, how will I change it's owner?
38. What is the difference between Hard Link and Soft Link?
39. How do you create a soft link?
40. How will you see what storage are mounted in the system?
41. What is the use of fstab?
42. What are the two default entries in fstab?
43. Assume you have mounted a disk in your system and you have deleted some files in it making enough space, but when you try to copy new files it gives you error "Not Enough space.", what could be the problem? (No the file system is correct ext4 and disk has correctly mounted as Read Write and not read only.)
44. Do you know what is crontab?
45. Write crontab entry for running a command on 6AM everyday.
46. How do you see CPU and RAM usage on your system?
47. When you run top command how will you check the Load on your system? (No not CPU or RAM usage)
48. What services have you used on AWS?
49. What is OPN sense?
50. What did you configure on your firewall?
51. How does the load balancing works?
52. How did you configure HA proxy?
53. Did you configure anything not in GUI?
54. Did you setup any load balancing inside your Kubernetes cluster?
55. I can see you have mentioned a certification in AWS in you resume, what all services did you learnt there?
56. Do you know what is AWS lambda? Have you heard of it?
57. How does a load balancer works in AWS?
58. What is Route 53 in AWS?
59. Assume you have your website running in EC2 instance and you have configured Route 53 for your website, then how does a user reaches your website after reaching Route 53 through

the DNS? What all AWS services will that connection encounter while going from Route 53 to your EC2 instance?

60. What is the firewall service in AWS?

61. Assume you have an EC2 instance what has allowed all connection including internet in both inbound and outbound rules, and you have a S3 storage, if you try to copy a file from your EC2 instance to that S3 will it give any error or will that be copied instantly?

Seems like they asked me quite a LOT of questions, in there I felt like some rapidfire round was going on, well their faces were laughing because I mostly gave wrong answers or just didn't knew answers to their questions at all. I hope you the reader gets some good out of all the torment I withstood.

TENERITY

Asked to Devyani:

1. What is a Pod in Kubernetes
2. your role in your project?
3. How did you create your Kubernetes cluster and configure its nodes?
4. Which type of EC2 instance did you use for your cluster, and why did you choose that particular instance type?
5. What disk size did you allocate to the instances while creating the nodes?
6. What were the minimum and maximum node sizes configured in your Kubernetes cluster's auto-scaling setup?
7. Can you explain the details of your Dockerfile and Kubernetes manifest (YAML) files?
8. How did you create your VPC, and can you explain each component in detail (CIDR block, subnets, route tables, private and public configuration)?
9. Can you provide the commands for the following?
To build a Docker image.
To deploy an application in Kubernetes.
To delete a Pod in Kubernetes.

10. Can you provide Linux commands for the following tasks?

Copying all files from one directory to another.

Checking memory status on a system.

11. Do you have knowledge of Jenkins or CI/CD pipelines? If yes, can you explain your understanding or experience?

12. Can you explain the different instructions used in your Dockerfile, such as FROM, COPY, RUN, WORKDIR, and CMD?

13. What are the main components of a Kubernetes cluster, and what roles do they play?

14. How would you resolve a conflict in Git if two developers try to perform a git pull at the same time?

15. Can you explain the role of a NAT Gateway and an Internet Gateway in your project's VPC setup? How natting works in VPC?
16. How many microservices were part of your application, and can you name them?
17. Apart from the services used in your project, which other AWS services are you familiar with?
18. How does AWS Lambda work?
19. How pods communicate with each other?
20. There is a table which shows you CPU utilization of different machines, I want to get the names of those machines with CPU utilization more than 70%, write a script?

Asked to Navnath Sangale

Everything about Linux , python code writing, kubernetes, docker swarm , AWS IAM , and all services we have learned in depth , and route 53,

Docker file writing,

What is user, role, and groups in IAM

Project ka ek ek point in depth

Scenario 1: you have python application (backend) take pen paper and explain how you will deploy it on VM or docker or using kubernetes, explain step by step in depth by using Jenkins , also write the pipeline or freestyle to deploy same web application

Scenario 2: you have VPC in VPC there are three subnets you want two subnets to communicate and 1 should not communicate other two , how you will do it

Scenario 3: how to create EC2 instance and tell me step by step and how you will SSH into it

Scenario 4: you're not able to SSH into EC2 what can be the cause , tell me the possibilities

Scenario 5: write a docker file to deploy web application and also use build: instruction write command to build also add tag to that particular image and how you will push it to DockerHub

Scenario 6:

I have one S3 bucket you have to connect it to EC2 , explain the process of connecting it

Bajaj Health

Asked to Rishi

- Q1. Tell me About Yourself?
- Q2. Which side you are interested (SOC or VAPT)
- Q3. Tell me about Your Project?
- Q4. Difference Between Encoding, Encryption And decryption
- Q5. Why you Need Log Rotation
- Q6. Difference between IDS & IPS
- Q7. What is SIEM?
- Q8. Which WAF you have used in your Project?
- Q9. How does the WAF Works?
- Q10. Difference Between window event vs window incident
- Q11. Difference between SAST VS DAST
- Q12. Which Firewall did you used?
- Q13. If a log is alerted on a SIEM tool like WAZU then what will you do?
- Q14. What is VA & PT
- Q15. if you are doing VA then how would you do it and what all will you check
- Q16. and based on VA what all PT will you do ?
- Q17. Tell me OWASP TOP 10
- Q18. What all attack's do you know and had Performed
- Q19. if a machine in your network has been infected or has been attacked what will you do ?
- Q20. What is Privacy Escalation

Asked to Sankalp

- 1. ports for dns, ftp, ssh, pop3, pop3s
- 2. explain how a mail transfers from my inbox to yours?
- 3. what you will do if a user is failing to login and his location shows from a different conrty.
- 4. What is compliance? what is dpdp?
- 5. What's the command to kill a proccess through process ID?
- 6. What's the command to see running processes?
- 7. What's the command to see open ports other than netstat?
- 8. What is lambda? whats difference between ec2 and lambda?
- 9. How can I deploy a piece of code on ec2?
- 10. Tell me the full form of OSI?
- 11. At which layer encryption happens? does it happens at transport layer?
- 12. difference between http and https?
- 13. Tell me all OWASP?
- 14. how does sql injection happen?
- 15. how does xss happen?
- 16. from where can we recover data and what kind of data can be recovered?

17. Assume in your organisation a system have become a part of ransomware attack now what will you do?
18. At which layer router works?
19. At which layer switches works?
20. What is ARP? what is ARP poisoning?
21. What is MITRE?
22. What is WAF?

Asked to Utkarsh

(Most of the questions are same as above ones so I am only adding distinct questions here)

ROUND 1

1. Tell me about yourself
2. Explain me about your project on the basis of what problem you encountered, how you solved it and the conclusion.
3. Which cybersecurity domains do you know about?
4. By which all commands we can see disk usage?
5. By which all commands we can see open ports?
6. How do you see all the running process in linux?
7. What is Kerberos?
8. What is [Golden ticket](#)?
9. What is [Silver ticket](#)?
10. Difference between encoding and encryption?
11. Are encryption and Hashing same?
12. Difference between Digital Certificate and Digital Signature
13. What is compliance?
14. What compliance do you know about?
15. What is PCI DSS?
16. What is GDPR?
17. Is GDPR an Indian compliance standard?
18. Do you know any Health related compliance standard?
19. How does DNS works, like when you enter www.somedomainname.com what happens in the background?
20. Assume a server has implemented an AES encryption with 2048 bit size key but it is observed that server performance is too slow so what alternatives can you suggest in this case?
21. On which OSI layer does HTTPS works?
22. Explain HTTPS handshake.
23. Explain TCP handshake.

There were few more questions but interviewer used some Short Forms which I've never heard about so I can't remember those.

ROUND 2

1. Tell me about yourself

2. Why did you choose cyber security?
 3. Which cybersecurity domains do you know about?
 4. Which role are you interested in, like SOC or VAPT? (VAPT was my [response](#))
 5. So assume you have a banking website, like SBI, according to VAPT thinking what security measures should be implemented? (Must answer atleast 10-15, I only did 8)
 6. Okay so now tell me what are you good at apart from academics?
 7. What job do your father do?
 8. Which version of SAP? [UNRELATED]
 9. From how much time are you here in Pune?
 10. Do you like the city?
 11. Do any of your relatives live here?
 12. Do you have any siblings?
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Bajaj Health Questions asked to Prathmesh

1. What is VAPT
2. What is the difference between va & pt
3. what is risk, threat, vulnerability
4. explain the osi model
5. explain all owasp top 10 (in detail)
6. sast vs dast
7. xss and types of xss in detail
8. how to check if any website is vulnerable to sql injection
9. what is xss and how to know if a website is vulnerable for xss
10. IDOR vs privilege escalation
11. types of hackers (all in detail)
12. encoding vs hashing vs encryption
13. example of encoding, hashing & encryption
14. symmetric & asymmetric key difference
15. what is sql injection? Explain all the ways to find whether the website is vulnerable to sql injection
16. how the xss and sql queries reflect (server & database)
17. what is file upload vulnerability
18. what is a payload
19. what is CIDR?
20. how to prevent sql and xss injection
21. Explain all the ways to exploit 2FA
22. How to prevent xss & SQL injection attacks (in detail)
23. What is Directory traversal attack? Explain In details
24. on which layer the router & switches work
25. On which osi layers DDoS occur? (App + Network)
26. if the attacker is using proxy chain to hide his IP, how to track him? Ans: i) by cooperating with ISP's. ii) by using canary tokens (a beacon payload which execute on attackers browser)

Everything was asked in detail

T-Systems Questions asked to **Nikhil D.**

(This questionnaire is typed by U.M. upon hearing crucial questions asked to his flatmate N.D.)

1. Tell me about yourself
2. What are all 4 default entries in fstab?
3. What does chage command do?
4. What are all 4 numbers in chage and in chmod command?
5. What is init command?
6. Explain all the runlevels.
7. What is GRUB?
8. What is difference between Kernel and GRUB?
9. Explain Linux boot process.
10. Which servers have you configured in Linux?
11. What are the configuration files in DHCP server?
12. What are the configuration files in DNS server?
13. How to see all the process with it's process ID?
14. How to force kill a process?
15. What -11 mean in KILL -11 command?
16. What is Master Boot record?
17. Difference between Legacy BIOS and UEFI BIOS?
18. What is a pod?
19. How is pod different from containers?
20. How does pod gets internet access?
21. What is the difference between CNI and Kube Proxy, explain their functions?
22. What does ADD do in a Dockerfile?
23. Difference between ADD and COPY in Dockerfile.
24. In AWS what is needed to configure in VPC?
25. Tell me all the functions on which pricing of AWS EC2 depends upon?

(There were more AWS cloud questions but due to their complexity and consisting of completely new terms those can't be remembered, rest of the questions were on personal life and hometown.)

T-Systems Questions asked to **Pawankumar**

1. In ps kill -9 <pid>, command what is -9?
2. What is a docker file, why is it used?
3. What is Docker Image?
4. Why docker, kubernetes is needed?
5. What is auto-scaling?
6. Instructions in docker file?

7. What is an entry point in docker?
 8. If your system is under memory pressure, a.How will you analyse it, and how will you resolve it?
 9. Why is linux preferred over windows for servers?
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T-Systems Questions asked to Keshav

- 1)Tell me about yourself,your hobbies and your educational qualification
 - 2)Explain workflow of your project?
 - 3)write all the commands of kubernetes you have used in your project.
 - 4)write any 10 linux//unix commands
 - 5)AWS services and their explanation?
 - 6)how are the pods and the nodes communicating with each other in your Kubernetes cluster?
 - 7)according to you which one is better Docker or Kubernetes and why?
 - 8)how did you insure security in your project?
 - 9)how can you justify your year gap which can be clearly seen in your resume :D
 - 10)tell me the command which is there to collect logs of the traffic which will be coming on your pods?:D(only question i could not answer but the interviewer already was convinced due to my previous answers by gods grace)
- Informal question=>how will you deal with a situation when a task is given to you ?will you say no....or what will be your approach?

HR INTERVIEW QUESTIONS

- 1)why are you interested in this role as a Cloud engineer ?
- 2)what do think about this field which you will be entering?
- 3)Which is your Aapp in your phone?whatever that app is what feature do you love about it
- 4)why do you feel underconfident about yourself?what are you doing consistently to make sure you feel confident about yourself?
- 5)currently what are you learning to make sure you are industry ready?
- 6)tell me about your weakness(i said i can't say no to people) XD
- 7)why can't you say no to people are you a people pleaser??? :D

Special mention to THE UTKARSH MISHRA(controller node) to make sure i perform well in this interview thanks to him and the almighty...